AWS EC2 – Auto Scaling Based on CPU Utilization (>70%)

Task:

Configure Auto Scaling so that new EC2 instances are automatically launched when CPU usage exceeds 70%.

Important Concept:

In Auto Scaling, we do NOT manually create an EC2 instance first.
Instead, AWS creates instances automatically using a Launch Template.

(a) Step 1: Create Launch Template

1. Login to AWS Management Console
   Open https://aws.amazon.com/ and sign in.
2. Open EC2 Dashboard
   Go to Services → EC2
3. Click **Launch Templates** (left panel)
4. Click **Create launch template**
5. Enter Details:
   - Launch template name: `AutoScaleTemplate`
   - Description: Template for Auto Scaling
6. Choose AMI:
   - Select **Amazon Linux 2** or **Ubuntu 22.04 LTS**
7. Select Instance Type:
   - `t2.micro` (Free Tier)
8. Key Pair:
   - Select existing key pair (example: `mykey.pem`)

9. Security Group:
    - Allow SSH (Port 22)
    - Allow HTTP (Port 80)
10. Click **Create launch template**

(b) Step 2: Create Auto Scaling Group

1. Go to EC2 Dashboard
2. Click **Auto Scaling Groups** (left panel)
3. Click **Create Auto Scaling group**
4. Enter Name:
    - Example: `MyAutoScalingGroup`
5. Select Launch Template:
    - Choose `AutoScaleTemplate`
6. Click **Next**
7. Select Network:
    - Choose VPC and subnets
8. Set Group Size:
    - Desired capacity: 1
    - Minimum capacity: 1
    - Maximum capacity: 3
9. Click **Next** → **Next** → **Create Auto Scaling Group**

(c) Step 3: Configure Scaling Policy (CPU > 70%)

1. Open created Auto Scaling Group
2. Go to **Automatic scaling** → **Add policy**
3. Choose Policy Type:
    - Target tracking scaling
4. Set Metric:
    - Average CPU Utilization
5. Set Target Value:
    - 70%
6. Click **Create Policy**

(d) Step 4: How Auto Scaling Works

- If CPU usage > 70%:
  - New EC2 instance is automatically launched
- If CPU usage decreases:
  - Extra instances are automatically terminated

(e) Step 5: Verify Auto Scaling

1. Go to EC2 → Instances
2. Monitor running instances
3. Use CloudWatch to check CPU usage
4. When load increases:
   - New instance is created automatically
5. When load decreases:
   - Instances are removed automatically

Result:

- Launch template created successfully
- Auto Scaling Group configured
- CPU-based scaling policy set at 70%
- EC2 instances automatically scale based on workload